

XIAOYAN LI

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👤 SUMMARY

Machine learning researcher with Ph.D. in Computational Mathematics and 9+ years of experience in deep learning, LLMs, and AI systems. Research focus on graph neural networks, computer vision, LLM reasoning, computational neuroscience, and brain connectomics. First-place finish in NeurIPS 2021 competition. Experience deploying ML systems on cloud platforms (AWS, GCP) and building scalable data pipelines. Interdisciplinary expertise spanning healthcare AI, cognitive science, environmental data science, and quantitative modeling.

🎓 EDUCATION

Michigan State University

July 2023

- Ph.D. in Computational Mathematics, Science & Engineering
- Ph.D. (Dual Major) in Civil & Environmental Engineering
- MS in Computer Science & Engineering

Carnegie Mellon University, MS in Civil & Environmental Engineering

Jan 2015

China Agricultural University, BS in Agricultural Engineering

Jun 2013

💼 EXPERIENCE

Postdoctoral Researcher, Department of Automation, Tsinghua University, Beijing

Mar 2024 – Mar 2026

- Led research at the intersection of computational neuroscience and machine learning for healthcare, focusing on multimodal medical data fusion, LLM reasoning, and deep learning model design.
- Developed CareAttenNet, a temporal attention-based deep learning framework for automated nursing activity recognition from wearable sensor data, achieving state-of-the-art performance. Published at IEEE BIBM 2025.
- Designed a multi-modal deep learning framework integrating image (CNN) and tabular clinical data for psoriasis diagnosis. Published at IEEE BIBM 2025.
- Proposed IACD, an iterative adversarial detection framework that mines blind spots of frozen detectors to find concealed targets (illicit poppy, UAV imagery). Published at ECCV 2026 (a top-three CV conference).
- First author of five papers accepted at IEEE BIBM 2026 on brain attention, dual-process reasoning in LLMs, theory of mind, and neural efficiency.
- Developed prompt tuning with RAG framework to enhance cognitive reasoning in LLMs, improving GPT-4 accuracy on the Criteria Cognitive Aptitude Test from 75.0% to 81.2%. Demonstrated that attention mechanisms are critical for long-term logical inference. Published at ACM RMELS Workshop (SenSys 2024).
- Investigated sparse routing mechanisms in foundation models through the lens of computational neuroscience, comparing biological and artificial intelligence architectures. (*Under review, Tsinghua Science and Technology*)
- Led as corresponding author fMRI-based study of cortical temporal hierarchy on ~65 hours of naturalistic movie data across 7 functional networks, showing LSTM models outperformed ridge regression by 47–57% uniformly. Accepted at CogSci 2026 (acceptance rate 44.6%).
- Built graph neural network models for protein-protein binding affinity prediction with multimodal feature fusion.

Postdoctoral Researcher, Michigan State University, East Lansing, MI

Jul 2023 – Mar 2024

- Developed AI-driven systems integrating deep learning with industrial manufacturing data, focusing on multimodal data modeling, real-time data cleaning, and intelligent optimization.
- Investigated model generalizability, interpretability, and robustness for production-level industrial data analysis.

Software Developer, Lucy Labs, New York, NY

Jan 2022 – Aug 2022

- Designed and implemented quantitative trading algorithms integrating statistical arbitrage and momentum strategies for cryptocurrency and equity markets.
- Built end-to-end automated data pipelines for trading data fetching, cleaning, and storage using Google Cloud BigQuery, processing millions of daily exchange records.

- Deployed ML model training workflows on AWS using Ray and Kubernetes for distributed hyperparameter tuning and backtesting at scale.

Research Intern, Xie's AI Lab, Michigan State University

May 2021 – May 2022

- Developed a Heterogeneous Graph Neural Network (GNN) to remove batch effects in high-dimensional single-cell sequencing data, incorporating gene pathway information in message passing.
- Built MultiDataset-MultiTask learning framework, reducing protein expression RMSE from 1.19 to 0.51 (57% improvement).

Project Leader, Water Contaminants Quantification, MSU (concurrent with Ph.D.)

May 2018 – Jul 2023

- Developed a Transformer-U-Net model to recognize and quantify tap water contaminant concentrations from cell phone images using transfer learning on 625 nanochromatography images and 4,200 EDS mappings. Reduced analysis cost from hundreds of dollars to \$1 and time from days to seconds.
- Project awarded \$6,700 funding from CEE department.

📖 SELECTED PUBLICATIONS

- Li X, Jiang C, Chen J, Yang R. "IACD: Iterative Adversarial Collaborative Detection via Dual-Perspective Blind Spot Discovery." *European Conference on Computer Vision (ECCV)*, 2026. (Top-3 CV conference)
- Jiang C, Ding X, Liu X, Wei J, Li X*. "The Temporal Efficiency Paradox: Fast Brains React, Slow Brains Understand." *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci)*, 2026. (*Corresponding author; Accepted)
- Li X, et al. "Do Foundation Models Think Like Efficient Brains? Comparing Sparse Routing Mechanisms in Biological and Artificial Intelligence." *Tsinghua Science and Technology*, 2026. (Under Review)
- Li X, et al. "Apparent Inefficiency, Hidden Optimality: Why the Brain's Multimodal Attention Does Not Track Feature Strength." *IEEE BIBM*, 2026. (Accepted)
- Li X, et al. "Do Large Language Models Think Fast and Slow? Eliciting Dual-Process Behavioral Signatures with Implications for Biomedical AI." *IEEE BIBM*, 2026. (Accepted)
- Li X, et al. "Attention Mechanisms for Theory of Mind: Weak Brain-Transformer Alignment Despite Behavioral Success." *IEEE BIBM*, 2026. (Accepted)
- Li X, et al. "Network Stability as the Hallmark of Neural Efficiency Under Cognitive Load." *IEEE BIBM*, 2026. (Accepted)
- Li X, et al. "Routing-Level Signatures of Cognitive Efficiency: A Multi-Task fMRI Null with a Pairwise-Correlation Caveat." *IEEE BIBM*, 2026. (Accepted)
- Li X, et al. "CareAttenNet: Deep Learning Framework with Temporal Attention for Automated Nursing Activity Recognition from Wearable Sensors." *IEEE BIBM*, 2025.
- Li X, et al. "A Multi-modal Deep Learning Framework for Psoriasis Diagnosis by Integrating Unaffected Fingernail Morphology and Clinical Indicators." *IEEE BIBM*, 2025.
- Li X, Jiang C. "Leveraging Prompt Tuning-Based Cognitive Attention to Enhance Logical Inference in Large Language Models." *ACM RMELS Workshop, SenSys*, 2024.
- Li X. "Attention as a Driving Force in Large Language Model Advancements." *Cognitive Computational Neuroscience Conference*, 2024.
- Li X, Sanderson AR, Allen SS, Lahr RH. "Tap water fingerprinting using a convolutional neural network built from images of the coffee-ring effect." *Analyst*, 2020; 145(4):1511-1523. (Cited: 14)
- Liu K, ..., Li X, et al. "Robust Synthesis of Targeting Glyco-Nanoparticles for Surface Enhanced Resonance Raman Based Image-Guided Tumor Surgery." *Small Science*, 2024. (Cited: 9)

🏆 AWARDS

1st Place, NeurIPS 2021 Single-cell Analysis Competition (Team: DANCE)

Nov 2021

- Developed the first GNN to predict RNA expression from chromatin accessibility, achieving the overall lowest RMSE of 0.2835 across all competing teams.

EPA Environmental Science Lahr Innovation Award

2017

⚙️ SKILLS

Languages: Python, C, C++, CUDA C++, Bash, SQL, R, HTML/CSS/JavaScript

ML/AI Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, PyTorch Geometric, Deep Graph Library, scikit-learn, XGBoost

LLM & GenAI: LLM pre-training (BabyLM), fine-tuning (LoRA/PEFT), RAG

Data & Cloud: AWS (S3, EC2, SageMaker), GCP (BigQuery), Docker, Kubernetes, Ray, MLflow, Git

Domains: Computer vision, GNNs, Transformers, time-series forecasting, sensor data analysis, statistical modeling (PCA, SVD, Bayesian methods, regression), multimodal data fusion